AES: open attack angles and a native discovery laboratory

17 October 2026. **Advisory research assessment and engineering work**, not an official goal transition, a security proof, or a claim to have broken AES. The implementation and first frozen synthetic suite are in `aburan28/cryptanalysis`. The native mirror is applied in `crypto` under current standalone authorization; its verification/delivery and separate scientific admission obligations are tracked in the problem record.

Assessment

The strongest research direction is to track **relations among several texts, their joint compatibility, and key-dependent constraints**, while building an instrument that can distinguish these mechanisms from data-generation and search artifacts. Counting active S-boxes or running a larger undisciplined search is insufficient. A useful binary needs a proposal engine, an exact cipher identity, a frozen observable, independent validation and complete cost records.

Recent primary literature gives concrete reasons to continue structural work. Shin et al. report a seven-round secret-key structural distinguisher with approximately `2^126.2` data, time and memory and 77.8% modeled success, with small-scale verification. Yan et al. report exchange/shift related-differential distinguishers through seven rounds. These results identify additional structure; their round counts and access models do not establish a practical attack on the 10/12/14-round standard ciphers. Both sources were inspected at abstract level, not independently reproduced. S03, S04

The cross-ratio preprint reports improved matching for a seven-round AES-128 meet-in-the-middle attack and reduced-round quartet distinguishers. It explicitly does not claim a threat to full AES. Our component experiment tests a known inversion identity and the AES output-affine layer; it does **not** reproduce that complete construction. Full algorithm retrieval remains a prerequisite to an attack-level reproduction. S05

The frontier is consequently open to better mathematics and engineering, with no basis here for predicting an imminent practical full-AES break. We preserve costly intermediate advantages as legitimate research outcomes: a reduced-round result, a memory reduction, or a model correction can matter without immediately recovering a practical key. Each needs its own evidence and accounting.

Coverage of attack directions

This is a broad taxonomy, not a proof that every conceivable method has been enumerated. `Open` means a specified research obligation remains; it does not mean a weakness is established. Priorities below are advisory to this tool project and do not reorder the existing official AES campaign queues.

Direction	Concrete open question / obstruction	First discriminating experiment	Access and evidence boundary
Differential trails and hulls	How much endpoint probability is missed by one trail or a capped beam, and which local paths are actually compatible?	Exact toy endpoint enumeration versus DDT-model/beam mass, then per-key AES endpoint tests	Chosen plaintext; local DDT products are not fixed-key probabilities or bounds on a hull
Related differences, exchange and joint quartets	Can a compact joint relation predict more than independent marginal events?	Count marginals, intersections and complete quartet predicates on a complete toy domain	Keep CP and adaptive CP distinct; published seven-round costs are not a local-run budget
Cross-ratio / algebraic matching	Which information must survive the real output-affine map and multi-byte mixing for a useful match?	Local inversion/affine ablations, then exact super-box witnesses	Component identity does not imply complete-round propagation; reject degenerate quartets
Boomerang / rectangle / retracing	Which middle-switch dependencies amplify, cancel or forbid the proposed quartet?	Exact BCT including its nonzero diagonal; fixed-delta returns against a consistent forward/ inverse PRP	Adaptive inverse access is a stronger model than CP; never choose delta from the observed ciphertext pair
Differential-linear and linear hulls	Do multi-round dependencies change the prediction from separate differential and linear scores?	Exact local DL tables and direct toy joint counts before combining scores	A correlation/model product requires its dependency assumptions to be checked
Integral / higher-order / division property	Which set summaries yield sound balance predictions, and which are merely permitted abstract trails?	Known three-round 256-text integral, one-member mutants, exact toy transition checks and next-round decay	A solver-allowed trail is not a balance certificate; a failed search is not security
Impossible differentials / zero correlation	Can exact state/key constraints rule out a new class beyond existing scoped exclusions?	SAT witness/UNSAT-certificate replay against complete toy enumeration	Zero observed events is not impossibility; existing one-word-in/one-word-out exclusions must not be rebranded as new
Key schedule / related-key / weak-key classes	Does a statistic depend on real expansion, a small key subset, or the data path alone?	Real schedule versus independent round keys, with per-key histograms; related-key panels separately	Related keys need a related-key ideal-cipher comparison and do not establish an ordinary single-key attack
DS-MitM / partial sums / bicliques	Can matching constraints lower complete time/data/memory costs?	Reproduce a named matching subproblem; count offline work, online work and large-table accesses separately	A cheaper subroutine is not a complete key-recovery improvement; biclique shortcuts remain enormously expensive
Algebraic / SAT / Gröbner / hybrid solving	Can exact value constraints or elimination exploit structure beyond a compact equation description?	Exhaustive S-box encoding equivalence, planted toy witnesses, solver counting and independently checked certificates	Timeout is not UNSAT; small-scale/known-bits results do not establish full-key AES scaling
Learned proposal/statistic models	Is there incremental information beyond classical histograms on untouched keys and structures?	Native key-disjoint data, simple baselines, shuffled labels, matched PRPs, then one untouched final test	Training/checkpoint selection and negative-generator artifacts can manufacture accuracy; charge all training and inference
Invariants, subspaces and alternative representations	Is the observable genuinely lossy and useful, with valid propagation under the actual operations?	Equal-observable collision pairs passed through each real layer; matched random-bijection/null ablations	Inversion is not the complete AES S-box; preserving values differs from propagating differences
Quantum search / structured quantum cryptanalysis	What changes under Q1 versus coherent Q2 access, with realistic circuit/depth/memory accounting?	Reproduce resource estimates on reversible S-box / structured-search subproblems	Separate from this classical binary and from practical quantum hardware; no numerical quantum claim is made here
Faults, leakage, white-box and protocol/mode failures	What additional access or misuse explains an observed compromise?	Separate synthetic implementation fixtures and explicit access models	These are important security tracks but do not by themselves demonstrate a mathematical core-cipher break

Sources for these families include the designers' wide-trail and plateau analyses, BCT and differential-linear dependency work, state-propagation/SAT, small-scale algebraic work, DS-MitM, related-key, yoyo and quantum analyses. S02, S07, S08, S09, S10, S11, S12, S06, S13, S16, S17.

What the first binary implements

The native Rust `aes-discover` crate implements AES-128/192/256 with explicit rounds and last-round MixColumns, exact DDT/LAT/BCT summaries, bounded value-level beam proposals, empirical endpoint discovery, hashed frozen validation, integrals, local quartet ablations, key-schedule differences, fixed-delta boomerang checks, and JSONL dataset export. Search considers byte and bit equalities of XOR projections of output differences. This is a first finite search family, not a generic solver for every taxonomy row.

The validator fixes a key across queries, uses new keys and plaintexts, excludes duplicate/reversed pairs, retains every key/event row and tests an independent lazy PRP. It uses conservative Chernoff bounds with finite-query coupling slack and family correction, rather than a sparse-cell normal approximation. A candidate must pass its frozen direction on every tested key, with the null family passing. No universal-key or optimality statement follows. The outer command card binds executable, sources, lockfile, protocol, exact commands and transcripts. Hashing an artifact does not independently certify its execution.

The first suite is a **separate engineering/measurement study**. It searches sixteen input differences per family and validates two events on eight new keys. Twelve predeclared families include ordinary full-round AES and random-control families, alongside the reduced-round positive controls. Other commands probe known integrals and component relationships. The native engine and frozen card are in the companion `cryptanalysis` PR; actual raw data and source bindings are published there. See the experiment adapter.

Why the instrument comes first

The source audit found issues that a new experiment should not inherit:

- Existing `milp.rs::enumerate_low_weight_trails` contains hard-coded small-round counts despite describing a search. It is not an exhaustive search certificate.
- Existing `differential.rs::propagate_one_round` marks an entire output column active if any input is active. For multiple active inputs, this is a possible activity visualization, not exact propagation. Two-byte cancellations exist; branch number five does not force all four outputs active in that case.
- Existing boomerang uniformity code skips `a == d`; the definition only excludes zero differences. A separate boomerang path sets output delta equal to the realized ciphertext difference and therefore returns for every permutation.
- Historical neural iterator code restarts a deterministic stream on each iterator creation, and repeatedly selects checkpoints using one evaluation set. Its independent-random-output negative class also differs from a shared PRP oracle. These are source findings, not a claim that every historical result has been experimentally refuted.

The new native path avoids these constructions. Historical files remain unchanged. KN-FIND-028/029 reinforce the same boundaries: MixColumns supplies a particular linear map, not every possible linear transformation; local value geometry is not difference propagation; actual exercised coverage and sibling nulls matter. No official conclusion is revised by this source audit.

Ranked follow-up experiments

The machine-readable cards in `hypotheses.json` contain the object, prediction, controls, falsification, resource envelope and acceptance criteria. Their novelty is **unverified** and `sota_delta` remains null.

1. Expand the native instrument panel: mutation/no-op controls, more null keys, independent implementation replay and counterexamples for every convention.
2. Enumerate a complete small-scale domain to compare endpoint hulls, beam proposals and per-key trail feasibility. Plateau behavior makes pooled-only claims particularly risky. S07, S08
3. Implement the **full** joint-quartet and related-difference predicates after retrieving their algorithms, then measure joint rather than multiplied marginal events. S03, S04
4. Connect local cross-ratio information to a precisely defined multi-byte matching object; expose affine-map and pole cases explicitly. S05
5. Add conditional boomerang / differential-linear switch checks and a certificate-producing SAT backend. S09, S10, S11
6. Extend integral certificates and schedule-isolating sibling controls, then attempt learned ranking only after the native classical baselines are sound. S15

These are proposals, not completed experiments. The existing three campaign queues retain their priorities. No scientific state changes or approvals are recorded here, and no held D-2 arm is launched.

Claim screening and remaining gaps

A credible new claim must identify the exact game and access model, full round count, unknown key bits, fixed versus key-averaged behavior, selected versus untouched data, complete probability or success criterion, preprocessing and per-target work, memory and memory accesses, and an independent reproducible check. A “faster than exhaustive search” headline alone does not establish practicality; a fixed-key trail is not a differential hull. Historical biclique results illustrate the distinction. S14

An August 2026 black-box preprint reports experiments with up to 80 unknown key bits when the other 48 are known, then extrapolates to the full-key case. We do not treat that extrapolation as an accepted full-AES break. Its full argument has not been independently reviewed here. S18

Open gaps are explicit: exact SAT/UNSAT certification, exhaustive toy hull accounting, full published quartet constructions, learned-model training, GPU search, controlled external performance comparisons, and official campaign admission are not implemented by this first release. A small negative search settles none of them. No new mathematical theorem or Lean proof is claimed.

Sources and provenance

`sources.json` records URLs, read scope, retrieval date and provenance. Most ePrint sources were inspected through their primary abstracts; their full PDFs returned HTTP 403 in this environment. The CiC DS-MitM PDF and the designers' plateau paper were retrieved and relevant sections inspected. An abstract-only read supports a report of the authors' claims, not an independent verification of their construction or costs.

The committed AES goal heads and relevant RQ/KN records were read at `crypto-autoresearcher@509d83b69a1f65230422e58bf4491cbbec1ad5da`, with source audits at `crypto@c6d64436` and `cryptanalysis@6346d31a`. The partial symmetric frontier was consulted without modifying it. No callable `search_knowledge` or `find_related` MCP was available; the sparse focused scan is not a complete corpus or novelty audit. Independent advisory and software reviews were used; they are not an official scientific review round.

Existing canonical curve graphs, IC scoreboards and performance leaderboards remain unchanged: this assessment adds no curve, IC measurement or speed claim. The editable research map distinguishes implemented instruments from proposed extensions. The report PDF includes the map and source links. A software merge or green tests do not promote a scientific hypothesis.

